Targeted Memory Reactivation During Slow-Wave Sleep

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Abstract. We tested whether replaying learning-associated sounds during slow-wave sleep strengthens declarative memory.

In 120 adults, targeted memory reactivation during slow-wave sleep improved recall of word pairs by 18% relative to a control night.

The benefit was largest for items that were weakly learned before sleep.

Results. Sleep spindle density in the two seconds after each cue predicted whether the cued item was remembered the next morning.
Reactivation delivered during REM sleep produced no measurable benefit.
These findings suggest that spindles gate the reactivation of hippocampal memory traces during slow-wave sleep.